

Member ID: _____

Time: _____

Rank: _____

Network Administration Using Cisco (315)

NATIONAL 2025

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. A library is upgrading its public Wi-Fi network to accommodate more users and provide faster internet access. They need a standard that supports both high speeds and a large number of simultaneous connections. Considering these requirements, which Wi-Fi standard should the library implement?
 - A. 802.11ac
 - B. 802.11ax
 - C. 802.11n
 - D. 802.11g
2. If a network administrator wants to allow only specific IP addresses to access a web server on a router, which type of ACL should be implemented?
 - A. Standard ACL
 - B. Extended ACL
 - C. Named ACL
 - D. Reflexive ACL
3. In a large organization, what is the primary benefit of implementing VLANs?
 - A. Improved network performance
 - B. Enhanced physical security
 - C. Simplified network management
 - D. Increased internet speed
4. In a network setup, what is the primary purpose of VLAN pruning?
 - A. To optimize bandwidth usage by restricting VLAN traffic
 - B. To encrypt all network traffic
 - C. To filter incoming traffic
 - D. To prioritize network traffic
5. In a network with multiple routers, if Router A's FastEthernet0/0 interface IP address is 192.168.1.1/24 and is connected to Router B's FastEthernet0/1 interface with IP address 192.168.1.2/24, which command would enable the OSPF dynamic routing protocol between the two routers?
 - A. router ospf 1
 - B. network 192.168.1.0
 - C. ip ospf 1 area 0
 - D. ospf 1 network 192.168.1.0 0.0.0.255

6. In dynamic routing, what is the process of routers agreeing on a consistent view of the network topology called?
 - A. Subnetting
 - B. Routing Table
 - C. Convergence
 - D. Routing Protocol
7. In network management, what is the purpose of setting up syslog monitoring?
 - A. To configure access control lists for network devices
 - B. To monitor and manage network devices remotely
 - C. To establish trunks between switches
 - D. To exchange routing information with other devices
8. Lucas is troubleshooting his slow home Wi-Fi network. He learns that the band he's using is overcrowded. Which band is likely to be less congested?
 - A. 2.4 GHz
 - B. 5 GHz
 - C. 11 GHz
 - D. 60 GHz
9. A network administrator configures a route that is used when no other route to a specific destination is available. What type of route is this in routing?
 - A. Direct Route
 - B. Static Route
 - C. Default Route
 - D. Dynamic Route
10. A network administrator is configuring FHRP to improve network reliability. Which feature allows for seamless failover between routers in case of primary router failure?
 - A. Shared virtual IP addresses
 - B. VLAN trunking
 - C. VLAN tagging
 - D. VLAN pruning
11. A network administrator is configuring FHRP to provide fault tolerance and load balancing for gateway addresses. Which protocol allows for automatic failover and load sharing capabilities among routers?
 - A. GLBP (Gateway Load Balancing Protocol)
 - B. NAT (Network Address Translation)
 - C. VPN (Virtual Private Network)
 - D. DHCP (Dynamic Host Configuration Protocol)

12. A network administrator is configuring IPv6 addresses for devices that require manual assignment of addresses. Which addressing method should be used for this requirement?
- A. EUI-64
 - B. SLAAC
 - C. DHCPv6
 - D. Static Address Assignment
13. A network administrator needs to configure a switch to allow specific VLANs to communicate with each other. Which feature should be enabled on the switch for inter-VLAN routing?
- A. VLAN trunking
 - B. VLAN tagging
 - C. VLAN aggregation
 - D. VLAN routing
14. A network administrator needs to group devices based on their functions to improve network performance and security. Which technology should be utilized for this purpose?
- A. VLAN
 - B. NAT
 - C. VPN
 - D. DHCP
15. A network administrator wants to control traffic flow based on Layer 3 information. Which feature should be configured on the router?
- A. NAT
 - B. ACL
 - C. Trunking
 - D. VTP
16. A network administrator wants to filter traffic based on IP addresses and port numbers. Which feature should be configured on the router?
- A. NAT
 - B. ACL
 - C. Trunking
 - D. VTP
17. A network administrator wants to limit the number of MAC addresses that can access a switch port. Which feature should be configured on the switch port?
- A. VLAN
 - B. Port Security
 - C. Trunking
 - D. VTP

18. A network administrator wants to manage VLANs across the network centrally. Which feature should be enabled for centralized VLAN management?
- A. VLAN
 - B. Port Security
 - C. Trunking
 - D. VTP
19. A network administrator wants to translate private IP addresses to public IP addresses for outgoing traffic. Which feature should be configured on the router?
- A. NAT
 - B. ACL
 - C. VLAN
 - D. OSPF
20. A network where every device is connected to a common central device such as a hub is an example of which network topology?
- A. Mesh Topology
 - B. Ring Topology
 - C. Star Topology
 - D. Bus Topology
21. A router is using the EIGRP routing protocol. What type of routing algorithm does EIGRP use?
- A. Link-State
 - B. Distance-Vector
 - C. Path-Vector
 - D. Hybrid
22. A router is using the OSPF routing protocol. What type of routing algorithm does OSPF use?
- A. Link-State
 - B. Distance-Vector
 - C. Path-Vector
 - D. Hybrid
23. An organization is experiencing performance issues due to excessive broadcast traffic. Which network feature can help mitigate this problem by segmenting the broadcast domain?
- A. VLAN
 - B. NAT
 - C. VPN
 - D. Proxy

24. An organization needs to assign a unique, site-specific address range for internal communication in an IPv6 network. Which type of address space should be utilized for this purpose?
- A. Global Unicast
 - B. Link-Local
 - C. Site-Local
 - D. Unique Local
25. An organization wants to implement a feature that dynamically assigns IPv6 addresses to devices as they connect to the network. Which mechanism should be used for this purpose?
- A. SLAAC
 - B. DHCPv6
 - C. Anycast
 - D. Dynamic Address Allocation
26. In a network environment, what is the primary purpose of implementing FHRP (First Hop Redundancy Protocol)?
- A. Load balancing
 - B. Redundancy and high availability
 - C. Data encryption
 - D. Quality of Service (QoS)
27. What is the function of Stateless Address Autoconfiguration (SLAAC) in IPv6 networks?
- A. To assign unique IPv6 addresses to hosts without a DHCP server
 - B. To encrypt data transmission over the network
 - C. To secure the router from unauthorized access
 - D. To prioritize network traffic based on specific criteria
28. What is the primary advantage of using VLANs to segment network traffic?
- A. Improved network security and performance
 - B. Reduced hardware costs
 - C. Slower data transmission
 - D. Limited scalability
29. What is the primary function of a switch in a network?
- A. To forward data packets between different networks
 - B. To connect devices within a local network
 - C. To amplify and regenerate signals
 - D. To determine the best path for data transmission
30. What is the primary function of a VLAN membership mode in network configuration?
- A. To assign ports to specific VLANs
 - B. To encrypt all network traffic
 - C. To filter outgoing traffic
 - D. To manage DHCP requests

31. What is the primary function of a VLAN trunking protocol in network communication?
- A. To transport multiple VLAN traffic over a single link
 - B. To encrypt all network traffic
 - C. To assign IP addresses to VLANs
 - D. To manage DHCP requests
32. What is the primary function of Dynamic Host Configuration Protocol (DHCP) in LAN design?
- A. To manage network traffic based on IP addresses
 - B. To enable communication between VLANs
 - C. To assign IP addresses automatically to devices on the network
 - D. To secure the router from unauthorized access
33. What is the primary purpose of VLAN tagging on Ethernet frames?
- A. To identify the VLAN membership of the frame
 - B. To encrypt all network traffic
 - C. To filter outgoing traffic
 - D. To prioritize network traffic
34. What is the role of SNMP (Simple Network Management Protocol) in network management?
- A. To secure switch ports from unauthorized access
 - B. To monitor and manage network devices remotely
 - C. To establish trunks between switches
 - D. To configure dynamic routing protocols
35. What is the significance of implementing Quality of Service (QoS) in network design?
- A. To secure the router from unauthorized access
 - B. To prioritize network traffic based on specific criteria
 - C. To manage VLANs on the router
 - D. To exchange routing information with other routers
36. What is the significance of the Multicast Listener Discovery (MLD) protocol in IPv6 networks?
- A. To manage network traffic based on IP addresses
 - B. To facilitate communication with multicast groups
 - C. To secure the router from unauthorized access
 - D. To prioritize network traffic based on specific criteria
37. What is the significance of using IPv6 Link-Local addresses in network communication?
- A. To communicate within the same link or subnet
 - B. To secure the router from unauthorized access
 - C. To manage network traffic based on IP addresses
 - D. To enable communication between different network segments

38. When configuring a Cisco switch port for a VoIP phone to connect to a separate voice VLAN, which command is used to assign the port to the voice VLAN?
- A. switchport voice vlan
 - B. voice vlan
 - C. switchport access vlan
 - D. vlan voice
39. When configuring ACLs for IPv4, what is the purpose of using extended ACLs compared to standard ACLs?
- A. To manage network traffic based on IP addresses and additional criteria
 - B. To secure the router from unauthorized access
 - C. To determine the best path for data transmission
 - D. To enable communication between VLANs
40. When configuring FHRP, what is the role of a virtual IP address?
- A. It provides a single gateway address for hosts
 - B. It encrypts all network traffic
 - C. It filters incoming traffic
 - D. It prioritizes network traffic
41. Which command is used to configure SSH access on a Cisco router for secure remote management?
- A. ssh enable
 - B. ip ssh version
 - C. transport input telnet
 - D. crypto key generate rsa
42. Which component is used to connect different network segments together?
- A. Switch
 - B. Router
 - C. Hub
 - D. Repeater
43. Which device operates at the Data Link layer of the OSI model and uses MAC addresses for communication within a local network?
- A. Router
 - B. Switch
 - C. Hub
 - D. Repeater

44. Which network component forwards data packets between different networks based on IP addresses in the packet headers?
- A. Switch
 - B. Router
 - C. Hub
 - D. Repeater
45. Which protocol operates at the Network layer of the OSI model and provides logical addressing for devices on the network?
- A. ARP
 - B. IP
 - C. TCP
 - D. ICMP
46. Which routing protocol uses the DUAL algorithm to calculate the best path to a destination network?
- A. RIP
 - B. OSPF
 - C. EIGRP
 - D. BGP
47. Which TCP/IP layer is responsible for logical addressing?
- A. Internet
 - B. Data Link
 - C. Transport
 - D. Application
48. You are troubleshooting a network issue and want to check the IP configuration on a Cisco router. Which command provides detailed information about the router's IP settings?
- A. show running-config
 - B. show ip interface
 - C. show interfaces
 - D. show ip route
49. You need to configure a static IPv4 route on a Cisco router to reach network 192.168.10.0/24 via the next-hop IP address 10.1.1.1. Which command accomplishes this task?
- A. ip route 192.168.10.0 255.255.255.0 10.1.1.1
 - B. ip route 0.0.0.0 0.0.0.0 192.168.10.0
 - C. ip route 10.1.1.1 255.255.255.0 192.168.10.0
 - D. ip route 192.168.10.0 0.0.0.255 10.1.1.1

50. You want to configure a default route for IPv6 traffic to go out of a specific interface on a Cisco router. Which command would you use?
- A. IPv6 route 2001::1/64 serial 0/1/0
 - B. IPv6 route ::1/0 serial 0/1/0
 - C. IPv6 route ::/0 serial 0/1/0
 - D. IPv6 route fe80::1/64 serial 0/1/0